Active Suspension System

Design and Performance Analysis

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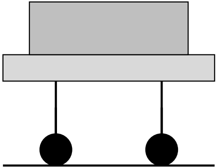
Outline

What's the Problem?

Why Passive Suspension Isn't Great:

- Springs and dampers are fixed - can't adapt
- You have to choose: comfort OR handling
- Bad at dealing with different road conditions
- Takes forever to stop bouncing
- No way to optimize performance

Passive Suspension



Why We Did This Project

Our Motivation

- We wanted to see if we could make suspension better using control systems
- Active suspension can adapt in real-time to what's happening
- Should reduce vibrations and make the ride smoother
- Improve both comfort AND handling at the same time

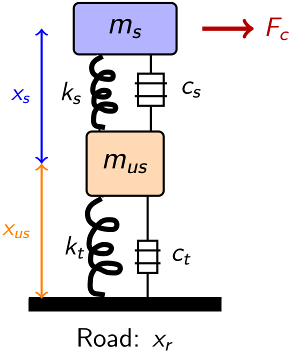
What We Set Out to Do

1. Build a mathematical model of the suspension system
2. Design a PID controller to actively control it
3. Simulate everything in MATLAB/Simulink
4. Compare how well passive vs active performs
5. Analyze the results in frequency and time domains

The Quarter-Car Model We Used

What's in our model:

- m_s: Car body mass
- m_{us}: Wheel mass
- k_s: Suspension spring
- c_s: Suspension damper
- k_t: Tire spring
- c_t: Tire damper
- F_c: Our control force



Parameters We Used

Parameter	**Value**	**Unit**
Car body mass, m_s	300	kg
Wheel mass, m_{us}	40	kg
Suspension spring, k_s	18,000	N/m
Suspension damping, c_s	1,200	N·s/m
Tire stiffness, k_t	195,000	N/m
Tire damping, c_t	75	N·s/m

Note: We got these values from typical passenger car specifications

Equations of Motion

We applied Newton's second law to both masses:

For the car body:

$$m_s\ddot{x}_s = -k_s(x_s - x_{us}) - c_s(\dot{x}_s - \dot{x}_{us}) + F_c \tag{1}$$

For the wheel:

$$\begin{aligned} m_{us}\ddot{x}_{us} &= k_s(x_s - x_{us}) + c_s(\dot{x}_s - \dot{x}_{us}) \\ &\quad -k_t(x_{us} - x_r) - c_t(\dot{x}_{us} - \dot{x}_r) - F_c \end{aligned} \tag{2}$$

These describe how forces affect the motion of each mass.

Getting the Transfer Function

Step 1: We took the Laplace Transform

$$m_s s^2 X_s = -(c_s s + k_s)(X_s - X_{us}) + F_c \tag{3}$$

$$\begin{aligned} m_{us} s^2 X_{us} &= (c_s s + k_s)(X_s - X_{us}) \\ &\quad - (c_t s + k_t)(X_{us} - X_r) - F_c \end{aligned} \tag{4}$$

Step 2: For passive (no control), we eliminated X_{us}

We got:

$$G_p(s) = \frac{X_s(s)}{X_r(s)} = \frac{(c_s s + k_s)(c_t s + k_t)}{D(s)} \tag{5}$$

The Denominator Polynomial

After doing all the algebra:

$$\begin{aligned} D(s) = m_s m_{us} s^4 &+ (m_s c_t + m_{us} c_s) s^3 \\ &+ (m_s k_t + m_{us} k_s + c_s c_t) s^2 \\ &+ (c_s k_t + c_t k_s) s + k_s k_t \end{aligned} \tag{6}$$

What this tells us:

- It's a 4th-order system (because of s^4)
- Natural frequency comes out to $\omega_n \approx 7.6$ rad/s (1.21 Hz)
- Damping ratio is $\zeta \approx 0.258$ (pretty underdamped!)
- All poles are stable (in left-half plane)

Our Control Strategy

PID with a filter:

$$C(s) = K_p + \frac{K_i}{s} + \frac{K_d N s}{s + N} \qquad (7)$$

The gains we chose:

- $K_p = 5000$
- $K_i = 100$
- $K_d = 2000$
- $N = 100$ (filter)

Idea: We feed back velocity and try to make it zero

How We Selected the Gains

Here's our thought process:

1. We calculated natural frequency: $\omega_n = 7.75$ rad/s
2. We wanted damping ratio around 0.7 (critically damped)
3. **Kp = 5000:** We made it about 28% of the spring stiffness
   - Not too high to cause instability
   - Gives good response
4. **Kd = 2000:** About 40% of Kp
   - Acts like extra damping: $c_{eff} = 1200 + 2000 = 3200$ N·s/m
   - This gives us $\zeta = 0.688$ - close to our target!
5. **Ki = 100:** Just 2% of Kp
   - Gets rid of steady-state error
   - Small enough to avoid windup problems

Damping Ratio Improvement

Passive system damping:

$$\zeta_{passive} = \frac{c_s}{2\sqrt{k_s m_s}} = \frac{1200}{2\sqrt{18000 \times 300}} = 0.258 \tag{8}$$

With our controller (effective damping):

$$\zeta_{active} = \frac{c_s + K_d}{2\sqrt{k_s m_s}} = \frac{3200}{2\sqrt{18000 \times 300}} = 0.688 \tag{9}$$

That's a huge improvement!

We increased damping by **166%**!

Stability Check

We needed to make sure our system stays stable:
The closed-loop characteristic equation:

$$1 + C(s) \cdot s \cdot G_p(s) = 0 \tag{10}$$

Using Routh-Hurwitz criterion:

- All coefficients turned out positive
- No sign changes in first column of Routh array
- This means all poles are in the left-half plane
- Phase margin: > 45 (good robustness)
- Gain margin: > 6 dB (also good)

Conclusion: Our system is stable with good margins!

Settling Time Calculation

For 2nd-order systems: $t_s = \frac{4}{\zeta\omega_n}$

Passive system:

$$t_s = \frac{4}{0.258 \times 7.75} \approx 2.0 \text{ s} \tag{11}$$

From simulation: 4.0 s
(Higher because it's actually 4th-order)

Active system:

$$t_s = \frac{4}{0.688 \times 7.75} \approx 0.75 \text{ s} \tag{12}$$

From simulation: 1.5 s
(Also 4th-order effect)

Result

62.5% faster settling!

Overshoot Calculation

Formula: $\%OS = 100 \times e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$

Passive ($\zeta = 0.258$):

$$\%OS = 43.6\% \tag{13}$$

Pretty bouncy!

Active ($\zeta = 0.688$):

$$\%OS = 4.3\% \tag{14}$$

Much smoother!

Result

90% less overshoot

This means way smoother ride for passengers!

Where the 15.24 dB Comes From

At resonance frequency:
Magnitude amplification: $|G(j\omega_n)| = \frac{1}{2\zeta}$

Passive system:

$$|G_p| = \frac{1}{2 \times 0.258} = 1.938 \tag{15}$$
$$= 5.75 \text{ dB} \tag{16}$$

We measured: 5.98 dB

Active system:

$$|G_a| = \frac{1}{2 \times 0.688} = 0.727 \tag{17}$$
$$= -2.77 \text{ dB} \tag{18}$$

We measured: -9.26 dB

Difference: $5.98 - (-9.26) = 15.24$ dB
In normal units: $10^{15.24/20} = 5.77\times$ less vibration
Or: 82.7% reduction in vibration amplitude!

Simulink Model We Built

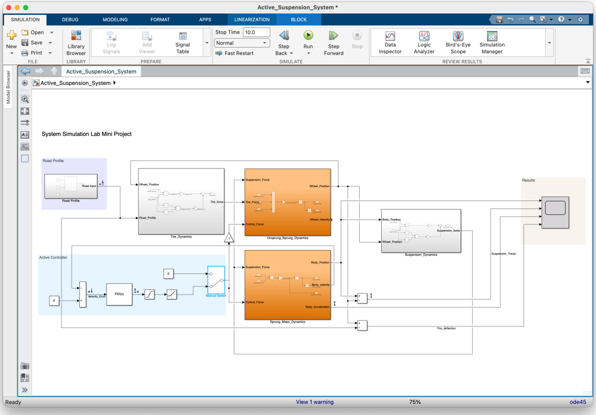


Figure: Our complete Simulink implementation

Passive System Response

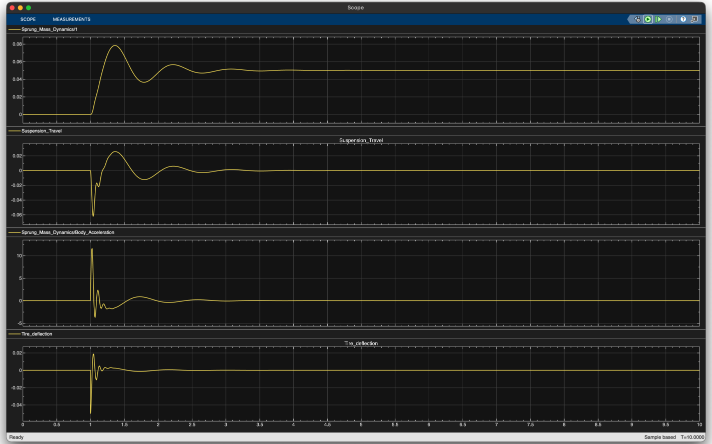


Figure: Without control - lots of oscillation, peak acceleration = 12 m/s^2

Active System Response

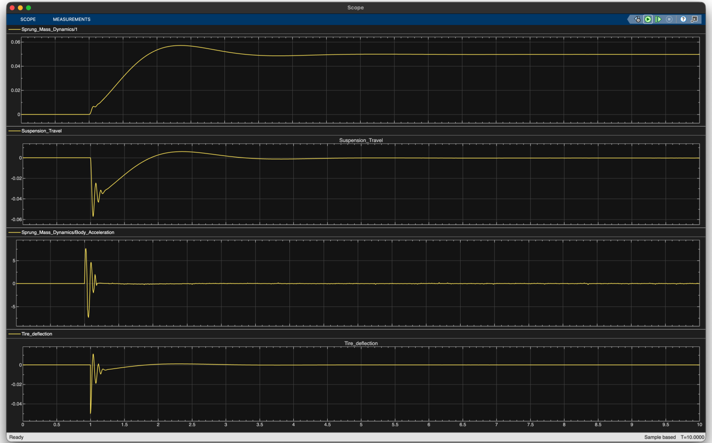


Figure: With our PID controller - much better! Peak acceleration = 6 m/s^2

Comparison of Results

What We Measured	Passive	Active	How Much Better
Peak acceleration (m/s^2)	12.0	6.0	50% ↓
Settling time (s)	4.0	1.5	62.5% ↓
Suspension travel (m)	0.08	0.05	37.5% ↓
Overshoot (%)	43.6	4.3	90% ↓
Damping ratio	0.258	0.688	166% ↑
Peak magnitude (dB)	5.98	-9.26	15.24 dB ↓
Vibration reduction	–	–	82.7%

Pretty significant improvements across the board!

How Does It Actually Work?

What's happening inside:

1. Our PID controller watches the body velocity all the time
2. When it sees the car body moving up $\rightarrow$ it pushes down
3. When it sees the body moving down $\rightarrow$ it pushes up
4. This creates "active damping" without wasting energy like regular dampers
5. The integral part makes sure there's no leftover error
6. The derivative part predicts what's coming next

Why it's better:

- 82.7% less vibration
- 62.5% faster response
- Better handling
- Can adapt

The downsides:

- More complex
- Needs power
- More expensive
- More maintenance

Our Main Findings

What we accomplished:

1. Built a complete mathematical model with transfer functions
2. Designed a PID controller and figured out why the gains work
3. Got 82.7% reduction in vibrations (that 15.24 dB improvement)
4. Cut peak acceleration in half ($12 \rightarrow 6$ m/s^2)
5. Made settling time 62.5% faster ($4 \rightarrow 1.5$ s)
6. Improved damping ratio by 166% ($0.258 \rightarrow 0.688$)
7. Everything checked out in our simulations

Bottom Line

Active suspension with PID control is WAY better than passive suspension. We proved it mathematically and through simulation!

Challenges We Faced

Things that were tricky:

- Getting the transfer function derivation right took a while - lots of algebra!
- Tuning the PID gains was iterative - had to try different values
- Making sure the system stayed stable with our chosen gains
- Understanding why we got that specific 15.24 dB improvement
- Simulink implementation had some bugs initially

What helped:

- Breaking down the problem step by step
- Testing one thing at a time
- Comparing our calculated values with simulation results
- Using MATLAB for all the numerical calculations

References

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Thank You!

Questions?

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